

Code description

A systematic engineering framework for robust medical resource allocation during public health emergencies considering secondary mortality

1. seir_simulation.py

Simulates the evolution of the COVID-19 epidemic using the Susceptible-Exposed-Infectious-Recovered (SEIR) epidemiological model.

2. population_weighted_Kmeans.py

Implements a population-weighted K-means clustering algorithm, where the population size is used as the weight of each representative demand point to aggregate them into a set of primary demand points.

3. medical_resource_allocation_robust_cplex_fuzzy_programming.py

Solves the main optimization model using CPLEX and a fuzzy programming approach to obtain a compromise solution. The script further compares the proposed model with its counterpart that ignores secondary mortality in the objective function, and performs sensitivity analyses on the parameters η , δ , and θ .

4. medical_resource_allocation_robust_cplex_epsilon.py

Solves the main optimization model using CPLEX and the epsilon-constraint method to generate a set of Pareto efficient solutions for the multi-objective optimization problem.

5. comparison_between_robust_deterministic.py

Compares the out-of-sample performance of the nominal-value-based deterministic formulation and the proposed robust optimization formulation.

6. comparison_allocation_strategy.py

Compares the proposed medical resource allocation strategy with two benchmark strategies, namely the Primary-first strategy and the Equal-ratio allocation strategy.